

Electrical Circuits

Lec 6: First Order Circuits

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Book

- **Fundamentals of Electric Circuits**
 - Charles K. Alexander
 - Matthew N. O. Sadiku
- 5th edition

First Order Circuit: The Source-Free RC Circuit

- A **first-order** circuit is characterized by a first-order differential equation.
- Consider a series combination of a resistor and an **initially charged capacitor**, as shown in Fig. 7.1.
- Our objective is to determine the circuit response, which, we assume to be the voltage $v(t)$ across the capacitor.
- Since the capacitor is initially charged, we can assume that at time $t=0$ the initial voltage is

$$v(0) = V_0 \quad (7.1)$$

with the corresponding value of the energy stored as

$$w(0) = \frac{1}{2} CV_0^2 \quad (7.2)$$

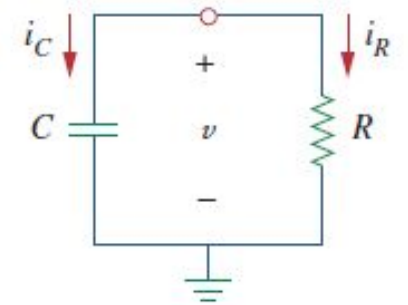


Figure 7.1
A source-free RC circuit.

Applying KCL at the top node of the circuit in Fig. 7.1 yields

$$i_C + i_R = 0 \quad (7.3)$$

By definition, $i_C = C dv/dt$ and $i_R = v/R$. Thus,

$$C \frac{dv}{dt} + \frac{v}{R} = 0 \quad (7.4a)$$

or

$$\frac{dv}{dt} + \frac{v}{RC} = 0 \quad (7.4b)$$

This is a *first-order differential equation*, since only the first derivative of v is involved. To solve it, we rearrange the terms as

$$\frac{dv}{v} = -\frac{1}{RC} dt \quad (7.5)$$

Integrating both sides, we get

$$\ln v = -\frac{t}{RC} + \ln A$$

where $\ln A$ is the integration constant. Thus,

$$\ln \frac{v}{A} = -\frac{t}{RC} \quad (7.6)$$

Taking powers of e produces

$$v(t) = Ae^{-t/RC}$$

But from the initial conditions, $v(0) = A = V_0$. Hence,

$$v(t) = V_0 e^{-t/RC} \quad (7.7)$$

- This shows that the voltage response of the RC circuit is an exponential decay of the initial voltage

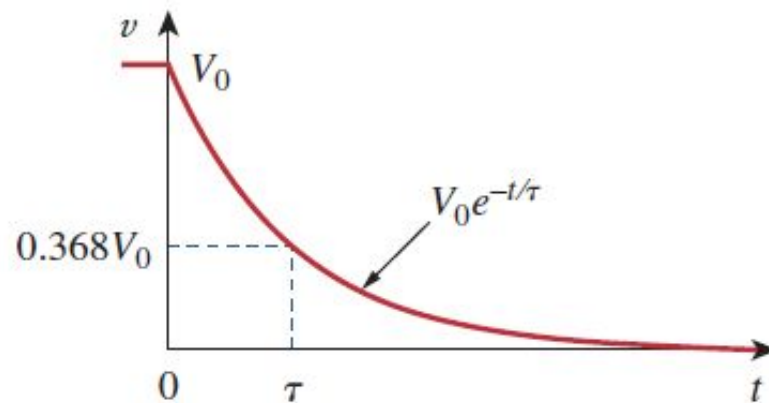


Figure 7.2

The voltage response of the RC circuit.

The time constant of a circuit is the time required for the response to decay to a factor of $1/e$ or 36.8 percent of its initial value

This implies that at $t = \tau$, Eq. (7.7) becomes

$$V_0 e^{-\tau/RC} = V_0 e^{-1} = 0.368 V_0$$

or

$$\tau = RC$$

(7.8)

In terms of the time constant, Eq. (7.7) can be written as

$$v(t) = V_0 e^{-t/\tau}$$

(7.9)

In Fig. 7.5, let $v_C(0) = 15$ V. Find v_C , v_x , and i_x for $t > 0$.

The $8\text{-}\Omega$ and $12\text{-}\Omega$ resistors in series can be combined to give a $20\text{-}\Omega$ resistor. This $20\text{-}\Omega$ resistor in parallel with the $5\text{-}\Omega$ resistor can be combined so that the equivalent resistance is

$$R_{\text{eq}} = \frac{20 \times 5}{20 + 5} = 4 \text{ } \Omega$$

Hence, the equivalent circuit is as shown in Fig. 7.6, which is analogous to Fig. 7.1. The time constant is

$$\tau = R_{\text{eq}}C = 4(0.1) = 0.4 \text{ s}$$

Thus,

$$v = v(0)e^{-t/\tau} = 15e^{-t/0.4} \text{ V}, \quad v_C = v = 15e^{-2.5t} \text{ V}$$

From Fig. 7.5, we can use voltage division to get v_x ; so

$$v_x = \frac{12}{12 + 8}v = 0.6(15e^{-2.5t}) = 9e^{-2.5t} \text{ V}$$

Finally,

$$i_x = \frac{v_x}{12} = 0.75e^{-2.5t} \text{ A}$$

Example 7.1

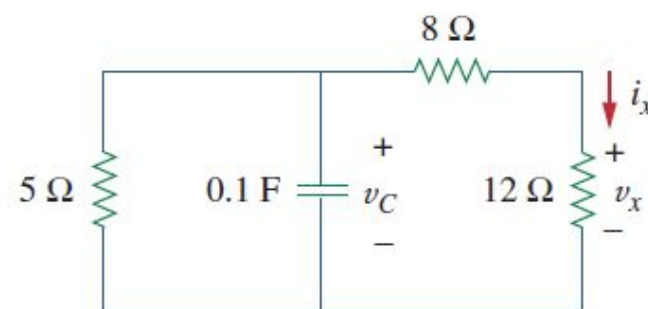


Figure 7.5
For Example 7.1.

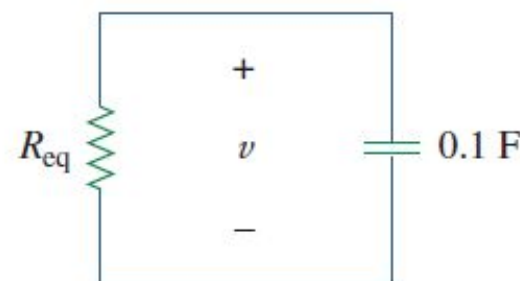


Figure 7.6
Equivalent circuit for the circuit in Fig. 7.5.

Practice Problem 7.1

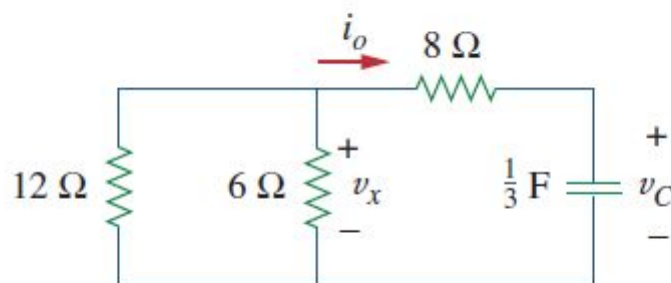


Figure 7.7

For Practice Prob. 7.1.

Refer to the circuit in Fig. 7.7. Let $v_C(0) = 60$ V. Determine v_C , v_x , and i_o for $t \geq 0$.

Answer: $60e^{-0.25t}$ V, $20e^{-0.25t}$ V, $-5e^{-0.25t}$ A.